

CSCI 5350

Advanced Topics in Game Theory

<http://course.cse.cuhk.edu.hk/~csci5350/>

1st Term, 2020/2021

Teacher: Ho-fung Leung
Time: T4-5, W3

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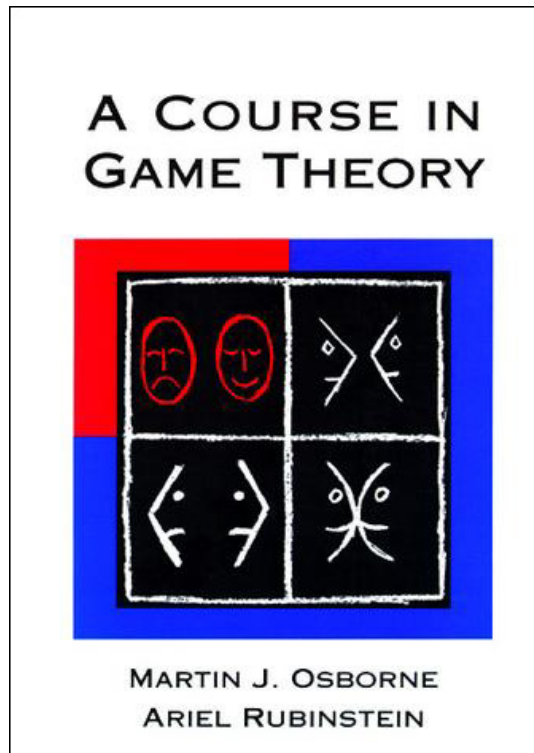
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Text Book



Osborne, M. J. and Rubinstein, A., 1994.
A Course in Game Theory. MIT Press.

Tentative Schedule:

We ek	Topics	Osborne & Rubin- stein	Remarks
1	Nash Equilibrium Zero-Sum Games	2.1 – 2.5	
2	Bayesian Games	2.6	
3	Mixed Strategies, Correlated Equilibrium and Evolutionary	3.1 – 3.4	Assign- ment 1
4	Extensive Games with Perfect Information	6.1	
5	Subgame Perfect Equilibrium	6.2 – 6.5	
6	Folk's Theorems	8.1 – 8.10	
7	Implementation Theory	10.1 – 10.5	
8	Extensive games with imperfect information	11.1	Assign- ment 2
9	Framing Effects	11.2 – 11.5	
10	Sequential Equilibrium	12.1 – 12.2	
11	Perfect Bayesian Equilibrium, Trembling Hand Perfect Equilibrium	12.3 – 12.5	
12	The Core	13.1 – 13.6	
13	Stable Sets and Shapley Value	14.1 – 14.2, 14.4	Assign- ment 3

Assessment Scheme:

Assignments: 45% (15% each)

Examination: 55%

- All assignments are written assignments.
- There is no programming assignment.
- There will be three assignments to be posted in week 3, week 8 and week 13.
- The final examination lasts for 2 hours.

The Prisoner's Dilemma



Player 2

Cooperate

Defect

Cooperate

Player 1

Defect

	<i>Cooperate</i>	<i>Defect</i>
<i>Cooperate</i>	freedom freedom	freedom & award imprisonment
<i>Defect</i>	imprisonment freedom & award	remission remission

If you are one of them, what will **YOU** do?

Strategic Games

Prisoner's Dilemma is a typical instance of **strategic games** (also known as '**normal form games**').

In general, in a strategic game:

- There is a finite set N of players.
- Player i has a set A_i of available actions.
- An **outcome** is an action profile: player 1 plays a_1 , player 2 plays a_2 , player 3 plays a_3 , and so on.
- For player i , some outcomes are better than ($\succsim_i$) other outcomes. (Each player thinks differently.)



Player 1

Cooperate

Defect

Player 2

Cooperate

Defect

	freedom	freedom & award
freedom		imprisonment
	imprisonment	remission
freedom & award		remission

$$N = \{1,2\}. \quad A_1 = \{C, D\}. \quad A_2 = \{C, D\}.$$

$$A = \{(C, C), (C, D), (D, C), (D, D)\}.$$

$$(D, C) \succeq_1 (C, C) \succeq_1 (D, D) \succeq_1 (C, D)$$

$$(C, D) \succeq_2 (C, C) \succeq_2 (D, D) \succeq_2 (D, C)$$

Strategic Games

DEFINITION. A **strategic game** $\langle N, (A_i), (\succeq_i) \rangle$ consists of

- a finite set N (the set of **players**)
- for each player $i \in N$ a nonempty set A_i (the set of **actions** available to player i)
- for each player $i \in N$ a preference relation $\succeq_i$ on $A = A_1 \times A_2 \times \dots \times A_n = \prod_{j \in N} A_j$ (the **preference relation** of player i on the set of **action profiles**).

Payoff Functions

In most cases the preference relation $\succeq_i$ can be represented by a **payoff function** (or *utility function*)

$$u_i: A \rightarrow \mathbb{R}$$

such that

$$u_i(a) \geq u_i(b) \text{ if and only if } a \succeq_i b.$$

In these cases, we can denote that game by $\langle N, (A_i), (u_i) \rangle$ rather than $\langle N, (A_i), (\succeq_i) \rangle$.



Player 1

Cooperate

Defect

Player 2

Cooperate

Defect

3, 3	0, 5
5, 0	1, 1

$$A = \{(C, C), (C, D), (D, C), (D, D)\}.$$

$$u_1((D, C)) = 5. \quad u_1((C, C)) = 3.$$

$$u_1((D, D)) = 1. \quad u_1((C, D)) = 0.$$

$$(D, C) \succeq_1 (C, C) \succeq_1 (D, D) \succeq_1 (C, C).$$

$$5 \geq 3 \geq 1 \geq 0$$

*Strategic games are also known as **bimatrix** games.*

Nash Equilibrium

$$(a_1^*, a_2^*, \dots, a_{k-1}^*, a_k^*, a_{k+1}^*, \dots, a_n^*) \in A$$

A **Nash equilibrium** is an action profile,

in which

every player thinks that he is playing the best strategy with respect to this profile.

NOTATIONS

$$a = (a_1, a_2, \dots, a_{k-1}, a_k, a_{k+1}, \dots, a_n)$$

$$a_{-k} = (a_1, a_2, \dots, a_{k-1}, \quad a_{k+1}, \dots, a_n)$$

$$(a_{-k}, x) = (a_1, a_2, \dots, a_{k-1}, x, a_{k+1}, \dots, a_n)$$

$$(a_{-k}, a_k) = (a_1, a_2, \dots, a_{k-1}, a_k, a_{k+1}, \dots, a_n)$$

$$(a_{-k}, a_k) = a$$

Nash Equilibrium

$$(a_1^*, a_2^*, \dots, a_{k-1}^*, a_k^*, a_{k+1}^*, \dots, a_n^*) \in A$$

In general, player k 's best strategy depends on other players' strategies. Assume a_k^* is the best strategy player k plays if other players do not change their strategies, that is, $(a_{-k}^*, a_k^*) \succeq_k (a_{-k}^*, a_k)$ for all $a_k \in A_k$.

If every player thinks this way, this action profile is in equilibrium, called the **Nash equilibrium**.

Nash Equilibrium

DEFINITION. A **Nash equilibrium** of a strategic game $\langle N, (A_i), (\succeq_i) \rangle$ is a profile $a^* \in A$ of actions with the property that for every player $i \in N$ we have

$$(a_{-i}^*, a_i^*) \succeq_i (a_{-i}^*, a_i) \text{ for all } a_i \in A_i.$$

Best Responses

We consider Nash equilibrium in terms of players' *best responses*. Consider

$$a_{-k} = (a_1, a_2, \dots, a_{k-1}, a_{k+1}, \dots, a_n)$$

Player k 's *best strategies* are

$$\{a_k \in A_k : (a_{-k}, a_k) \succeq_k (a_{-k}, a'_k) \text{ for all } a'_k \in A_k\}.$$

Player k 's *best-response function* B_k :

$$B_k(a_{-k}) = \{a_k \in A_k : (a_{-k}, a_k) \succeq_k (a_{-k}, a'_k) \text{ for all } a'_k \in A_k\}.$$

Consider a Nash equilibrium

$$(a_1^*, a_2^*, \dots, a_n^*) \in A.$$

By definition, each player is playing his best response.
Therefore, we have

$$a_i^* \in B_i(a_{-i}^*) \text{ for all } i \in N.$$

Example: The Prisoner's Dilemma



		Player 2	
		<i>Cooperate</i>	<i>Defect</i>
Player 1	<i>Cooperate</i>	3, 3	0, 5
	<i>Defect</i>	5, 0	1, 1

Where are the Nash Equilibria?

Example: Battle of the Sexes

(Also known as 'Bach or Stravinsky')

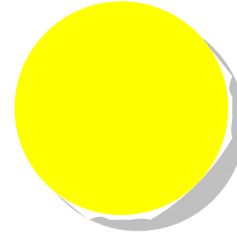
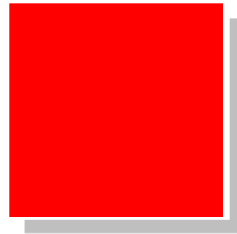


		Wife	
		<i>Boxing</i>	<i>Opera</i>
Husband	<i>Boxing</i>	2, 1	0, 0
	<i>Opera</i>	0, 0	1, 2

Where are the Nash Equilibria?

Example: Coordination Game

(Also known as 'Mozart or Mahler')



		Player 2	
		<i>Square</i>	<i>Circle</i>
Player 1	<i>Square</i>	2, 2	0, 0
	<i>Circle</i>	0, 0	1, 1

Where are the Nash Equilibria?

Example: Hawk-Dove Game

(Also known as 'Chicken Game')



		Bird 2	
		<i>Dove</i>	<i>Hawk</i>
Bird 1	<i>Dove</i>	3, 3	1, 4
	<i>Hawk</i>	4, 1	0, 0

Where are the Nash Equilibria?

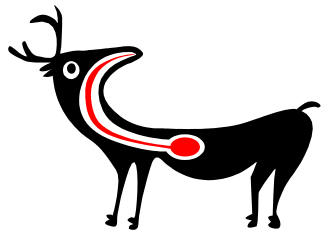
Example: Matching Pennies



		Player 2	
		<i>Head</i>	<i>Tail</i>
Player 1	<i>Head</i>	1, -1	-1, 1
	<i>Tail</i>	-1, 1	1, -1

Where are the Nash Equilibria?

Example: Stag Hunt



Player 2

Player 1

Stag

Hare

Stag

Hare

5, 5	0, 2
2, 0	1, 1

Where are the Nash Equilibria?

Class Discussion: First-Price Sealed-Bid Auctions

Player						
Valuations (v_i)	\$10m > \$7m > \$6m > \$3m > \$1m > 0					

Player with the lowest index wins if more than one player submits the highest bid.

Class Discussion

Player											
											
Valuations	v_1	$>$	v_2	$>$	v_3	$>$	v_4	$>$	v_5	$>$	0
Submitted Bid	b_1		b_2		b_3		b_4		b_5		

$$b_i \in A_i = [0, \infty).$$

$$u_i(b_{-i}, b_i) = \begin{cases} v_i - b_i & \text{Player } i \text{ wins} \\ 0 & \text{Otherwise} \end{cases}.$$

Q: What are the Nash equilibria?

Player											
Valuations (v_i)	v_1	$>$	v_2	$>$	v_3	$>$	v_4	$>$	v_5	$>$	0
Submitted Bid	b_1		b_2		b_3		b_4		b_5		

Check whether the profile $(b_1, b_2, \dots, b_n)$ is a Nash equilibrium:

- $b_1 \in [v_2, v_1]$,
- $b_j \leq b_1$ for all $j \neq 1$, and
- $b_j = b_1$ for some $j \neq 1$.

(Comment: b_1 has to be the highest.)

Are there any other Nash equilibria?

Class Discussion: Second-Price Sealed-Bid Auctions

Player					
Valuations (v_i)	v_1	v_2	v_3	v_4	v_5
Submitted Bid	b_1	b_2	b_3	b_4	b_5

$$A_i = [0, \infty). \quad u_i(b_{-i}, b_i) = \begin{cases} v_i - b_j & \text{Player } i \text{ wins} \\ 0 & \text{Otherwise} \end{cases}.$$

Player i wins if b_i is the highest bid.

(Note: b_j is the highest of the other players' bids).

Q: What are the Nash equilibria?

Player					
Valuations (v_i)	v_1	v_2	v_3	v_4	v_5
Submitted Bid	b_1	b_2	b_3	b_4	b_5

$v_1 > v_2 > v_3 > v_4 > v_5 > 0$

Hint:

Consider $(v_1, \dots, v_n)$, that is, $b_i = v_i$.

- $\max_{j \neq i} b_j \geq v_i$?
- $\max_{j \neq i} b_j < v_i$?

Q: Is it better for $b_i \neq v_i$?

!Q: Is there a Nash equilibrium in which player 3 wins the item?

There is a Nash equilibrium in which player 3 wins:

Player					
Valuations (v_i)	10	7	6	3	1
Submitted Bid	5	0	11	0	0

In general, any outcomes that satisfy the following conditions are Nash equilibria in which player j wins:

- $b_1 < v_j$.
- $b_j > v_1$.
- $b_i = 0$ for $i \notin \{1, j\}$.

Symmetric Games

A strategic game $\langle \{1,2\}, (A_i), (u_i) \rangle$ is **symmetric** if $A_1 = A_2$, and $(a_1, a_2) \succeq_1 (b_1, b_2)$ if and only if $(a_2, a_1) \succeq_2 (b_2, b_1)$ if for all $a \in A$ and $b \in A$.

Bird 2

		<i>Dove</i>	<i>Hawk</i>
Bird 1	<i>Dove</i>	3, 3	1, 4
	<i>Hawk</i>	4, 1	0, 0

$$(H, D) \succeq_1 (D, D) \leftrightarrow (D, H) \succeq_2 (D, D)$$

$$(D, H) \succeq_1 (H, H) \leftrightarrow (H, D) \succeq_2 (H, H)$$

Is there a symmetric N.E. (a^*, a^*) ?

Symmetric Games

A strategic game $\langle \{1,2\}, (A_i), (u_i) \rangle$ is **symmetric** if $A_1 = A_2$, and $(a_1, a_2) \succeq_1 (b_1, b_2)$ if and only if $(a_2, a_1) \succeq_2 (b_2, b_1)$ if for all $a \in A$ and $b \in A$.

	L	M
T	1, 1	0, 0
B	0, 0	0, 0

Is there a symmetric N.E. (a^*, a^*) ?

Symmetric Games

A strategic game $\langle \{1,2\}, (A_i), (u_i) \rangle$ is **symmetric** if $A_1 = A_2$, and $(a_1, a_2) \succeq_1 (b_1, b_2)$ if and only if $(a_2, a_1) \succeq_2 (b_2, b_1)$ if for all $a \in A$ and $b \in A$.

	<i>Left</i>	<i>Right</i>
<i>Left</i>	1, 1	0, 0
<i>Right</i>	0, 0	1, 1

Is there a symmetric N.E. (a^*, a^*) ?

Symmetric Games

A strategic game $\langle \{1,2\}, (A_i), (u_i) \rangle$ is **symmetric** if $A_1 = A_2$, and $(a_1, a_2) \succeq_1 (b_1, b_2)$ if and only if $(a_2, a_1) \succeq_2 (b_2, b_1)$ if for all $a \in A$ and $b \in A$.

	X	Y
X	0, 0	1, 1
Y	1, 1	0, 0

Is there a symmetric N.E. (a^*, a^*) ?

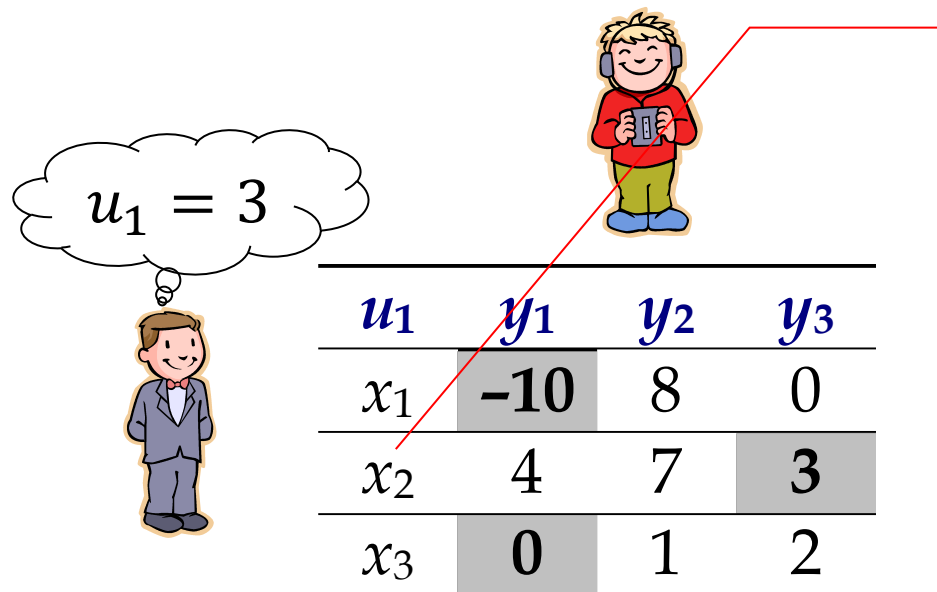
Strictly Competitive Games

DEFINITION. A strategic game $\langle \{1,2\}, (A_i), (\succeq_i) \rangle$ is **strictly competitive** if for any $a \in A$ and $b \in A$ we have $a \succeq_1 b$ if and only if $b \succeq_2 a$.

A strictly competitive game is sometimes called *zerosum* when the two players' payoffs can be represented by u_1 and u_2 with $u_1 + u_2 = 0$.

Maxminimisation

A player *maxminimises* if he chooses an action on the assumption that his opponent always hurts him as much as possible.



This action is called a *maxminimiser* for player 1.

u_1	y_1	y_2	y_3
x_1	-10	8	0
x_2	4	7	3
x_3	0	1	2

Maxminimiser Actions

DEFINITION. Let $\langle \{1,2\}, (A_i), (u_i) \rangle$ be a strictly competitive strategic game. The action $x^* \in A_1$ is a **maxminimiser for player 1** if

$$\min_{y \in A_2} u_1(x^*, y) \geq \min_{y \in A_2} u_1(x, y) \text{ for all } x \in A_1.$$



u_1	y_1	y_2	y_3
x_1	-10	8	0
$x^* = x_2$	4	7	3
x_3	0	1	2

Maxminimiser Actions

A maxminimiser maximises the payoff that the player can *guarantee*. A maxminimiser for player 1 solves the problem $\max_{x \in A_1} \min_{y \in A_2} u_1(x, y)$, that is,

$$x^* = \arg \max_{x \in A_1} \min_{y \in A_2} u_1(x, y)$$



u_1	y_1	y_2	y_3
x_1	-10	8	0
x^*	4	7	3
x_3	0	1	2

In the following, without loss of generality, we assume that

$$u_2 = -u_1.$$

What if Player 2 also performs maximinisation?

$$y^* = \arg \max_{y \in A_2} \min_{x \in A_1} u_2(x, y)$$



	u_2	y_1	y_2	y^*
x_1		10	-8	0
x_2		-4	-7	-3
x_3		0	-1	-2



$u_2 = -3$



	u_1	y_1	y_2	y^*
x_1		-10	8	0
x_2		4	7	3
x_3		0	1	2

	u_2	y_1	y_2	y^*		$u_2 = -3$
	x_1	10	-8	0		
	x_2	-4	-7	-3		
	x_3	0	-1	-2		

	u_1	y_1	y_2	y^*		$u_1 = 3$
	x_1	-10	8	0		
	x_2	4	7	3		
	x_3	0	1	2		

The maxminimisation of player 2's payoff is equivalent to the minmaximisation of player 1's payoff.

$$\max_{y \in A_2} \min_{x \in A_1} u_2(x, y) = - \min_{y \in A_2} \max_{x \in A_1} u_1(x, y)$$

(Note: $u_2(x, y) = -u_1(x, y)$)

Equivalence of Maxminimisation of u_2 and Minmaximisation of u_1

LEMMA. Let $\langle \{1,2\}, (A_i), (u_i) \rangle$ be a strictly competitive strategic game. Then $\max_{y \in A_2} \min_{x \in A_1} u_2(x, y) = \min_{y \in A_2} \max_{x \in A_1} u_1(x, y)$. Further, $y \in A_2$ solves the problem $\max_{y \in A_2} \min_{x \in A_1} u_2(x, y)$ if and only if it solves the problem $\min_{y \in A_2} \max_{x \in A_1} u_1(x, y)$.

Nash Equilibrium and Maxminimisation



u_2	y_1	y_2	y^*
x_1	10	-8	0
x_2	-4	-7	-3
x_3	0	-1	-2



u_1	y_1	y_2	y^*
x_1	-10	8	0
x_2	4	7	3
x_3	0	1	2

In a Nash equilibrium (x^*, y^*) , both players are maxminimising.



u_2	y_1	y_2	y^*
x_1	10	-8	0
x_2	-4	-7	-3
x_3	0	-1	-2



u_1	y_1	y_2	y^*
x_1	-10	8	0
x_2	4	7	3
x_3	0	1	2

Consider a Nash equilibrium (x^*, y^*) .

- First, $u_2(x^*, y^*) \geq u_2(x^*, y)$ for all $y \in A_2$, or,
 $u_1(x^*, y^*) \leq u_1(x^*, y)$ for all $y \in A_2$.

Hence $u_1(x^*, y^*) = \min_{y \in A_2} u_1(x^*, y) \leq \max_{x \in A_1} \min_{y \in A_2} u_1(x, y)$.

- Similarly, $u_1(x^*, y^*) \geq u_1(x, y^*)$ for all $x \in A_1$.

Hence $u_1(x^*, y^*) \geq \min_{y \in A_2} u_1(x, y)$ for all $x \in A_1$.

Hence $u_1(x^*, y^*) \geq \max_{x \in A_1} \min_{y \in A_2} u_1(x, y)$ for all $x \in A_1$.

$$\therefore u_1(x^*, y^*) = \max_{x \in A_1} \min_{y \in A_2} u_1(x, y).$$

In Strictly Competitive Games, Nash Equilibrium Implies Maxminimisation

PROPOSITION. Let $G = \langle \{1,2\}, (A_i), (u_i) \rangle$ be a **strictly competitive** strategic game.

If (x^*, y^*) is a Nash equilibrium of G then x^* is a maximiniser for player 1 and y^* is a maximiniser for player 2.

This is an important result to remember:

In a **strictly competitive** strategic game, if (x^*, y^*) is a Nash equilibrium, then both players are maxminimising in (x^*, y^*) !

And we now go to the next important result...



u_2	y_1	y_2	y^*
x_1	10	-8	0
x_2	-4	-7	-3
x_3	0	-1	-2



u_1	y_1	y_2	y^*
x_1	-10	8	0
x_2	4	7	3
x_3	0	1	2

Now we know that in a Nash equilibrium (x^*, y^*) ,

$$u_1(x^*, y^*) = \max_{x \in A_1} \min_{y \in A_2} u_1(x, y), \text{ and}$$

$$u_2(x^*, y^*) = \max_{y \in A_2} \min_{x \in A_1} u_2(x, y),$$

$$u_1(x^*, y^*) = -u_2(x^*, y^*)$$

$$u_1(x^*, y^*) = -\max_{y \in A_2} \min_{x \in A_1} u_2(x, y) = \min_{y \in A_2} \max_{x \in A_1} u_1(x, y).$$

$$\text{Therefore, } \max_{x \in A_1} \min_{y \in A_2} u_1(x, y) = \min_{y \in A_2} \max_{x \in A_1} u_1(x, y).$$

Value of Strictly Competitive Games



u_2	y_1	y_2	y^*
x_1	10	-8	0
x_2	-4	-7	-3
x_3	0	-1	-2



$u_2 = -3$



u_1	y_1	y_2	y^*
x_1	-10	8	0
x_2	4	7	3
x_3	0	1	2



$u_1 = 3$

$$u_1(x^*, y^*) = \max_{x \in A_1} \min_{y \in A_2} u_1(x, y) = 3$$

$$u_2(x^*, y^*) = \max_{y \in A_2} \min_{x \in A_1} u_2(x, y) = -3$$

Uniqueness of Payoff in Nash Equilibrium

PROPOSITION. Let $G = \langle \{1,2\}, (A_i), (u_i) \rangle$ be a **strictly competitive** strategic game.

If (x^*, y^*) is a Nash equilibrium of G then $\max_x \min_y u_1(x, y) = \min_y \max_x u_1(x, y) = u_1(x^*, y^*) = v^*$ and thus all Nash equilibria of G yield the same payoff v^* (called the '**value** of the game').

Class Discussion

IN THE GAME MATCHING PENNIES, WE HAVE

$$\max_x \min_y u_1(x, y) = ?$$

$$\min_y \max_x u_1(x, y) = ?$$

They are not equal?

		Player 2	
		<i>Head</i>	<i>Tail</i>
Player 1		1, -1	-1, 1
		-1, 1	1, -1
		<i>Head</i>	<i>Tail</i>

Class Discussion

IN THE GAME OF BATTLE OF THE SEXES, WHICH IS NOT STRICTLY COMPETITIVE, WE DO NOT HAVE

$$\max_{x \in A_1} \min_{y \in A_2} u_1(x, y) = \min_{y \in A_2} \max_{x \in A_1} u_1(x, y)$$



Husband

Boxing
Opera

Wife

Boxing

Opera

2, 1	0, 0
0, 0	1, 2

This is an important result to remember:

In a **strictly competitive** strategic game, all Nash equilibria yield the same payoff v^* !

And we now go to the last important result...

Given a strictly competitive game, let

$$\max_x \min_y u_1(x, y) = \min_y \max_x u_1(x, y) = v^*, \quad \text{hence}$$

$$\max_y \min_x u_2(x, y) = -\min_y \max_x u_1(x, y) = -v^*.$$

Let x^* and y^* be maxminimisers. Hence $u_1(x^*, y) \geq v^*$ for all $y \in A_2$, and $u_2(x, y^*) \geq -v^*$ for all $x \in A_1$. Let $y = y^*$, we obtain $u_1(x^*, y^*) = v^*$; and let $x = x^*$, we obtain $u_2(x^*, y^*) = -v^*$.

We conclude that (x^*, y^*) is a Nash equilibrium of G .

Uniqueness of Maxminimised Payoff Implies Nash Equilibria Being Maxminimisation

PROPOSITION. Let $G = \langle \{1,2\}, (A_i), (u_i) \rangle$ be a **strictly competitive** strategic game.

If $\max_x \min_y u_1(x, y) = \min_y \max_x u_1(x, y)$, x^* is a maximinimiser for player 1, and y^* is a maximinimiser for player 2, then (x^*, y^*) is a Nash equilibrium of G .

- If (x^*, y^*) is a Nash equilibrium of G then x^* is a maximinimiser for player 1 and y^* is a maximinimiser for player 2.
- If $\max_x \min_y u_1(x, y) = \min_y \max_x u_1(x, y)$, x^* is a maximinimiser for player 1, and y^* is a maximinimiser for player 2, then (x^*, y^*) is a Nash equilibrium of G .

This is an important result to remember:

In a **strictly competitive** strategic game, the Nash equilibria are *interchangeable*: if (x, y) and (x', y') are equilibria, then so are (x, y') and (x', y) .

PROPOSITION. Let $G = \langle \{1,2\}, (A_i), (u_i) \rangle$ be a **strictly competitive** strategic game.

- a. If (x^*, y^*) is a Nash equilibrium of G then x^* is a maximiniser for player 1 and y^* is a maximiniser for player 2.
- b. If (x^*, y^*) is a Nash equilibrium of G then $\max_x \min_y u_1(x, y) = \min_y \max_x u_1(x, y) = u_1(x^*, y^*) = v^*$ and thus all Nash equilibria of G yield the same payoff v^* (called the '**value** of the game').
- c. If $\max_x \min_y u_1(x, y) = \min_y \max_x u_1(x, y)$, x^* is a maximiniser for player 1, and y^* is a maximiniser for player 2, then (x^*, y^*) is a Nash equilibrium of G .

Class Discussion



	u_2	y_1	y_2	y_3
x_1		10	-8	0
x_2		-4	-7	-3
x_3		8	-3	-9



$u_2 = ?$



	u_1	y_1	y_2	y_3
x_1		-10	8	0
x_2		4	7	3
x_3		8	3	9



$u_1 = ?$

$$\max_x \min_y u_1(x, y) = ? \quad 3$$

$$\min_y \max_x u_1(x, y) = ? \quad 7$$